Posterior for Poisson Likelihood with Gamma Prior

Goal. Derive the posterior distribution for the Poisson rate parameter λ for a Poisson likelihood and Gamma prior on λ.

Likelihood. We observe N independent counts $x = \{x_1, x_2, \dots, x_N\}$, and we model

$$X_i \mid \lambda \;\sim\; \text{Poisson}(\lambda) \quad \text{independently for } i = 1, \dots, N.$$

The Poisson pmf is

$$\Pr(X_i = x_i \mid \lambda) \;=\; \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}, \qquad x_i \in \{0, 1, 2, \dots\},\; \lambda > 0.$$

Under independence, the *likelihood* of λ given the data is

$$f(x \mid \lambda) \;=\; \prod_{i=1}^{N} \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} \;=\; \left[\frac{1}{\prod_{i=1}^{N} x_i!}\right] e^{-N\lambda}\, \lambda^{\sum_{i=1}^{N} x_i}.$$

Prior. Place a Gamma prior on λ with shape $\alpha^* > 0$ and rate $\beta^* > 0$.

$$\lambda \;\sim\; \text{Gamma}(\alpha^*, \beta^*), \qquad f(\lambda) \;=\; \frac{(\beta^*)^{\alpha^*}}{\Gamma(\alpha^*)}\, \lambda^{\alpha^*-1} e^{-\beta^*\lambda}, \quad \lambda > 0.$$

Task. Find the posterior distribution $f(\lambda \mid x)$, identifying it by name and parameters. (Hint: *The Gamma distribution is conjugate to the Poisson likelihood.*)

Explaining the Rejection Algorithm

Task. In your own words, explain the logic of the *rejection algorithm* to someone who has never seen it before. Assume:

- The target distribution has support on $[0, 1]$ and is known up to a proportionality constant; denote its (possibly unnormalized) density by $f(z)$ for $z \in [0, 1]$.
- We use a *uniform* proposal on $[0, 1]$, i.e., propose $Z \sim \text{Uniform}(0, 1)$.
- We have an *envelope constant* $M > 0$ satisfying $M \geq \sup_{z\in[0,1]} f(z)$.

Explain why the algorithm works and what the output represents. You may use sentences or bullet points, but explain with words rather than equations. And explain *why* it works rather than what it does.

Solutions

Solution to Question 1

Step 1: Write down the pieces explicitly.

From the pmf and independence,

$$\begin{aligned} f(x \mid \lambda) &= \prod_{i=1}^{N} \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} = \left[\frac{1}{\prod_{i=1}^{N} x_i!}\right] \left(\prod_{i=1}^{N} \lambda^{x_i}\right) \left(\prod_{i=1}^{N} e^{-\lambda}\right) \\ &= \underbrace{\left[\frac{1}{\prod_{i=1}^{N} x_i!}\right]}_{\text{constant in } \lambda} \lambda^{\sum_{i=1}^{N} x_i}\, e^{-N\lambda}. \end{aligned}$$

The Gamma prior density (shape–rate) is

$$f(\lambda) = \underbrace{\frac{(\beta^*)^{\alpha^*}}{\Gamma(\alpha^*)}}_{\text{constant in } \lambda} \lambda^{\alpha^*-1} e^{-\beta^* \lambda}, \qquad \lambda > 0.$$

Step 2: Multiply likelihood and prior (Bayes up to proportionality).

Ignoring constants that do not depend on λ,

$$\begin{aligned} f(\lambda \mid x) &\propto f(x \mid \lambda)\, f(\lambda) \\ &\propto \left(\lambda^{\sum_{i=1}^{N} x_i} e^{-N\lambda}\right) \left(\lambda^{\alpha^*-1} e^{-\beta^* \lambda}\right). \end{aligned}$$

Step 3: Collect like terms carefully (same base λ and same base e).

For powers of λ:

$$\lambda^{\sum_{i=1}^{N} x_i} \times \lambda^{\alpha^*-1} = \lambda^{\left(\sum_{i=1}^{N} x_i + \alpha^* - 1\right)}.$$

For exponentials:

$$e^{-N\lambda} \times e^{-\beta^* \lambda} = e^{-(N+\beta^*)\lambda}.$$

Therefore,

$$f(\lambda \mid x) \propto \lambda^{\left(\alpha^* + \sum_{i=1}^{N} x_i\right) - 1}\, e^{-(\beta^* + N)\lambda}.$$

Step 4: Name the updated parameters and recognize the family.

Define

$$\alpha' = \alpha^* + \sum_{i=1}^{N} x_i, \qquad \beta' = \beta^* + N.$$

Then

$$f(\lambda \mid x) \propto \lambda^{\alpha'-1} e^{-\beta' \lambda},$$

which is the *kernel* of a Gamma(α', β') density.

Step 5: Restore the normalizing constant (to make it a proper density).

The normalized Gamma density with shape α' and rate β' is

$$f(\lambda \mid x) = \frac{(\beta')^{\alpha'}}{\Gamma(\alpha')}\, \lambda^{\alpha'-1} e^{-\beta'\lambda}, \qquad \lambda > 0.$$

Hence the posterior is

$$\boxed{\lambda \mid x \sim \text{Gamma}\left(\alpha^* + \sum_{i=1}^{N} x_i,\ \beta^* + N\right).}$$

Why this makes sense (quick checks).

- *Conjugacy:* Gamma prior + Poisson likelihood $\Rightarrow$ Gamma posterior.
- *Learning from data:* More events (large $\sum x_i$) increase the posterior shape; more observations (N) increase the rate, shrinking toward stable estimates.

Solution to Question 2

Paragraph explanation Think of the target distribution as a wavy "height" curve drawn above the interval $[0, 1]$, and imagine a flat roof set high enough to sit above that entire curve. We first spread candidate points evenly across the floor from 0 to 1 (uniform proposals), which means every location is tried equally often. Then we keep each candidate with a chance that depends only on how tall the target curve is at that spot relative to the roof: where the curve is tall, candidates are kept often; where it is short, they are kept rarely.

This "shape-based thinning" turns an even carpet of proposals into a pile of kept points whose density mirrors the curve itself. The unknown normalizing constant of the target never matters, because we only compare relative heights; stretching or shrinking the entire curve by a common factor just changes how many proposals are kept overall, not *where* they are kept. The roof guarantees the keep–discard chance is valid everywhere and also determines efficiency (a lower roof, just above the peak, wastes fewer proposals).

The accepted z's are thus independent draws that behave as if they were generated directly from the target distribution, ready for summaries like means, quantiles, and credible intervals.